Answer

L 12 : PROBABILITIES (PART III)

Expected Value

	1	2	3	4	5	6
Probability of getting	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Expected Value

$$=1\times\frac{1}{6}+2\times\frac{1}{6}+3\times\frac{1}{6}+4\times\frac{1}{6}+5\times\frac{1}{6}+6\times\frac{1}{6}$$

$$= 3.5$$

Let's get started!

1. Probability of getting a \$10 note = $\frac{1}{4}$

   Probability of getting a \$20 note = $\frac{1}{4}$

   Probability of getting a \$50 note = $\frac{1}{2}$

   Find the expected value.

   Expected value

   $= \frac{1}{4}\times 20 + \frac{1}{4}\times 20 + \frac{1}{2}\times 50$

   $= \$32.5$

2. P (10 marks) = 20%
   P (20 marks) = 35%
   P (30 marks) = 45%
   Find the expected mark.

   Expected Value
   = 22.5

3. P (50 marks) = 80%
   P (100 marks) = 15%
   P (800 marks) = 5%
   Find the expected mark.

   Expected Value
   = 95

4. Macy takes part in a drawing competition. It is known that $\$3\,000$, $\$1\,000$ and $\$500$ will be awarded for the champion, 1st runner-up and 2nd runner-up respectively. If the probabilities that Macy becomes the champion, 1st runner-up and 2nd runner-up are 0.1, 0.2 and 0.3 respectively, find the expected value of her award.

The expected value of her award

$= \$(3\,000 \times 0.1 + 1\,000 \times 0.2 + 500 \times 0.3)$

$= \underline{\underline{\$650}}$

5. A wallet contains two $\$10$ notes, five $\$20$ notes and three $\$100$ notes. If a note is drawn from the wallet randomly, what is the expected value of the note drawn?

(a) Find the probabilities of drawing a $\$10$ note.

(b) Fin the expected value of the draw.

$P(\$10 \text{ drawn}) = \frac{2}{10} = \frac{1}{5}$

$P(\$20 \text{ drawn}) = \frac{5}{10} = \frac{1}{2}$

$P(\$100 \text{ drawn}) = \frac{3}{10}$

$\therefore$ The expected value of the note drawn

$= \$\left(10 \times \frac{1}{5} + 20 \times \frac{1}{2} + 100 \times \frac{3}{10}\right)$

$= \underline{\underline{\$42}}$

6. In a game, a player has to draw a marble randomly from a bag which contains 1 blue marble, 1 red marble and 1 green marble. The points the player can get are shown below:

Marble	Blue	Red	Green
Points	12	-18	9

Find the expected value of the points got by the player.

Expected value of the points

$= (12) \times \frac{1}{3} + (-18) \times \left(\frac{1}{3}\right) + (9) \times \left(\frac{1}{3}\right)$

$= 4 - 6 + 3$

$= \underline{\underline{1}}$

7. In a game, a player has to draw a card from 52 playing cards.

Card	A	2	3	4	5	6	7	8	9	10	J/Q/K
Points	+11	+5	+5	+5	+5	-5	-5	-5	-5	10	10

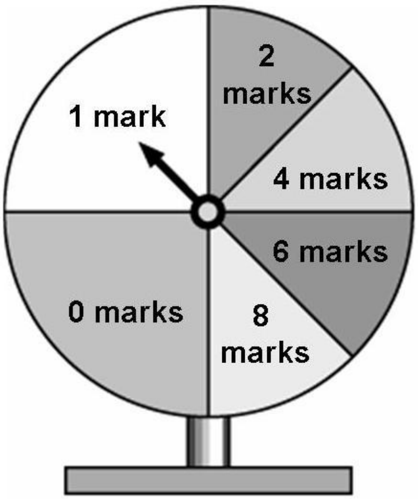
Find the expected value of the points got by the player.

Expected value

$= 11 \times \frac{4}{52} + (+5) \times \frac{16}{52} + (-5) \times \frac{16}{52} + (+10) \times \frac{16}{52}$

$= \frac{51}{13}$ (or 3.92 or $3\frac{12}{13}$)

8. The figure shows a lucky wheel. The semi-circle on the left of the wheel is divided into two equal sectors while the semi-circle on the right is divided into four equal sectors. Find the expected value of the marks obtained in spinning the lucky wheel once.



The following table lists all the possible outcomes and their corresponding theoretical probabilities:

Marks	0	1	2	4	6	8
Probability	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

$\therefore$ Expected value of the marks

$$= 0\times\frac{1}{4}+1\times\frac{1}{4}+2\times\frac{1}{8}+4\times\frac{1}{8}+6\times\frac{1}{8}+8\times\frac{1}{8}$$

$$= \underline{\underline{2.75}}$$

9. Entrance Fee for lucky draw: \$200
P (\$50) = 70%   P (\$150) = 20%   P (\$1000) = 10%
Calculate the expected value and decide whether it is worth to play the lucky draw.

Calculate the expected value:

Compare the values:

Conclusion:

Expected Value
= \$165
< \$200
No, it is not worth to play the lucky draw.

10. Maggie is planning to travel aboard. She had caught a cold and suffered from endemic disease in her previous trips, the medical fee was \$ 300 and \$ 1 000 respectively. Therefore, she is thinking of buying a travelling insurance of \$ 200. It is estimated that the probability of catching a cold and suffering from endemic disease are 0.2 and 0.1 respectively. Ignoring other factors, do you think that Maggie should buy the travelling insurance?

Expected medical fee
$= \$(300 \times 0.2 + 1\,000 \times 0.1)$
$= \$160$
Since the expected medical fee is less than the insurance premium, Maggie should not buy the travelling insurance.

11. An organization issues 50 000 lucky draw tickets. The prizes are as follows:

Prize item	Number of prizes	Prize
1st prize	1	\$ 100 000
2nd prize	1	\$ 50 000
3rd prize	3	\$ 10 000
Consolation prize	5	\$ 1 000

(a) Find the expected value of the prize of each of the lucky draw tickets.

(b) If Jacky buys a lucky draw ticket of \$ 20, does he gain or suffer a loss on average? Explain your answer.

(a) $P(\text{1st prize}) = \dfrac{1}{50\ 000}$

$P(\text{2nd prize}) = \dfrac{1}{50\ 000}$

$P(\text{3rd prize}) = \dfrac{3}{50\ 000}$

$P(\text{Consolation prize}) = \dfrac{5}{50\ 000}$

$= \dfrac{1}{10\ 000}$

Expected value of the prize of each of the lucky draw tickets

$$= \$\left(100\,000 \times \frac{1}{50\,000} + 50\,000 \times \frac{1}{50\,000} + 10\,000 \times \frac{3}{50\,000} + 1000 \times \frac{1}{10\,000}\right)$$

$$= \$(2 + 1 + 0.6 + 0.1)$$

$$= \underline{\underline{\$3.7}}$$

(b) Since the expected value of the prize of each of the lucky draw tickets is less than the price of the ticket, he will suffer a loss on average.

12. In a game, a player draws a ball at random from a box. In the box, there are 50 balls of the same size and each ball is marked with a distinct number from 1 to 50 inclusively. If the ball drawn is marked with one of the following numbers 8, 18, 28, 38 or 48, a \$ 300 will be awarded.

(a) Find the expected value of the amount of money a player gets in one game.

(b) If a player has to pay \$ 40 for playing the game once, is the game favourable to the player?

(a) $P(\$300) = \dfrac{5}{50} = \dfrac{1}{10}$

$P(\$0) = \dfrac{50-5}{50} = \dfrac{9}{10}$

Expected value of the amount of money

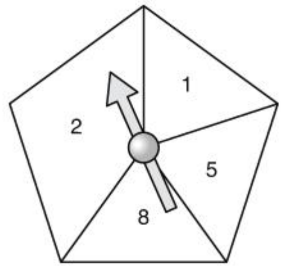
$$= \$\left(300 \times \frac{1}{10} + 0 \times \frac{9}{10}\right)$$

$$= \underline{\underline{\$30}}$$

(b) $\because$ Expected value of the amount of money a player get (\$30) is smaller than the cost of playing the game (\$40).

$\therefore$ The game is not favourable to the player.

13. The figure shows a lucky wheel in the shape of a regular pentagon. David spins the lucky wheel and gets the points marked on the region where the pointer stops.



(a) Find the probability that David gets
   (i) 5 points, (ii) 2 points.

(b) Find the expected value of the points David gets in one spin.

(c) David will receive a special prize if he gets 10 points or more in two spins. Given that David gets 8 points in the first spin, find the probability that he receives the special prize.

(a) **(i)** $\dfrac{\text{Area of the region with 5 points}}{\text{Area of the pentagon}} = \dfrac{1}{5}$

$\therefore \quad P(5\text{ points}) = \underline{\underline{\dfrac{1}{5}}}$

(ii) $\dfrac{\text{Area of the region with 2 points}}{\text{Area of the pentagon}} = \dfrac{2}{5}$

$\therefore \quad P(2\text{ points}) = \underline{\underline{\dfrac{2}{5}}}$

(b)

Point(s)	1	2	5	8
Probability	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{1}{5}$

$\therefore$ Expected value of the points

$= 1\times\dfrac{1}{5} + 2\times\dfrac{2}{5} + 5\times\dfrac{1}{5} + 8\times\dfrac{1}{5}$

$= \underline{\underline{3.6}}$

(c) For a special price, David has to get 2, 5 or 8 points in the second spin.

$\dfrac{\text{Area of the regions with 2, 5 or 8 points}}{\text{Area of the pentagon}} = \dfrac{4}{5}$

$\therefore \quad P(\text{special prize}) = \underline{\underline{\dfrac{4}{5}}}$

14. William plays a game. First he draws a ball from a bag containing three balls labelled 1, 2, and 3. If ball 3 is drawn in the first part, then he wins \$ 30. The ball drawn is put back and one ball is drawn from the bag again. However he will lose \$ 30 if the second ball is 3 too. If the ball drawn (not 3) is the same as in the first draw, William can still get \$ 12; otherwise, he can't get any money.

(a) Draw a table to list out all the possible outcomes when he draws two balls from the bag with replacement.

(b) Hence, find the expected amount of money William can win from the game.

(c) If it takes William \$ 10 to play the game, do you think the game is favourable to him?

2nd outcome

1st outcome	1	2	3
1	1,1	1,2	1,3
2	2,1	2,2	2,3
3	3,1	3,2	3,3

(b) $P(1,1) = \frac{1}{9}$

$P(2,2) = \frac{1}{9}$

$P(3,1/3,2) = \frac{2}{9}$

$P(3,3) = \frac{1}{9}$

Expected value

$= \frac{1}{9}\times 12 + \frac{1}{9}\times 12 + \frac{2}{9}\times 30 + \frac{1}{9}\times(-30)$

$= \$6$

(c) $\because$ $6 < 10$

$\therefore$ Not favour.

Give it a try

If we roll the dice twice, what is the expected sum of values?

Expected Value for the first roll

= 3.5

Expected Value for the second roll

= 3.5

Sum of the expected values

= 3.5 × 2 = 7

	1	2	3	4	5	6
Probability of getting	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Expected Value

$$= 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6}$$

$$= 3.5$$

You have learnt more! Go on!

15. Gary tosses two coins separately. He will be rewarded 50 marks if he gets a tail but deducted 20 marks if he gets a head. Find the expected total mark after the two tosses.

Expected total mark
$= [50 \times 0.5 + (-20) \times 0.5] \times 2$
$= 30$

16. There are 10 cards with number '0' to '9' respectively. Mary draws a card out of ten for three times separately and she will get the score with the same value as the cards she has got. Find the expected total score.

Expected total score
$= [(0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9) \times 0.1] \times 3$
$= 13.5$

17. There are 30 light bulbs in a batch. The probability that a light bulb is faulty is 0.03. Estimate the number of faulty light bulbs in 5 batches of light bulbs.

5 batches of light bulbs $= 5 \times 30 = 150$ light bulbs.

Expected number of faulty light bulbs $= 150 \times 0.03 = 4.5$

18. The probability that a newly born baby to be heavier than 8 pounds is 0.15. Doris has just given birth to 4 babies, find the expected number of Doris's babies with weight less than or equal to 8 pounds.

P(less than or equal to 8 pounds) $= 1 - 0.15 = 0.85$

Expected number $= 4 \times 0.85 = 3.4$

19. In a multiple-choice test paper, each question has three options A, B and C. Only one of the options is correct. Tom does not know how to answer two questions in the test, so he selects the answers randomly.

(a) List all the possible outcomes in a table. (Use T to stand for correct and F stand for wrong.)

Answer of second question

Answer of first question		T	F	F
	T			
	F			
	F			

(b) (i) Find the probability that the two answers chosen are correct.

(ii) Find the probability that only one of the two answers is correct.

(c) 2 marks will be rewarded for a correct answer while 0 mark will be given for incorrect answer. Find Tom's expected total score.

(a)

Answer of second question

Answer of first question		T	F	F
	T	T T	T F	T F
	F	F T	F F	F F
	F	F T	F F	F F

(b)(i) P (Two answers chosen are correct) $= \dfrac{1}{9}$ (ii) P (Only one of the answers is correct) $= \dfrac{4}{9}$

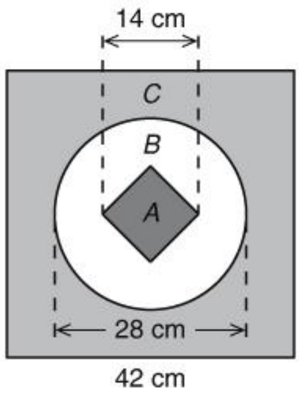
(c) Tom's expected total score $= 4 \times \frac{1}{9} + 2 \times \frac{4}{9} = 1.33$ or $(2 \times \frac{1}{3}) \times 2 = 1.33$

Final Boss!

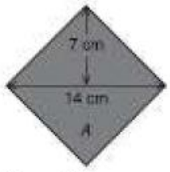
20. The figure shows a square dartboard with a circle and a square drawn inside. It is given that 9 marks will be awarded for hitting zone A, 3 marks will be awarded for hitting zone B and 2 marks will be deducted for hitting zone C. Tom throws a dart onto the dartboard at random. (Take $\pi = \frac{22}{7}$.)

(a) Find the probabilities of hitting each zone on the dartboard.

(b) Find the expected value of the marks Tom gets in one throw.



(a) Area of the dartboard $= 42^2\text{ cm}^2 = 1764\text{ cm}^2$



$\because$ Zone A is a square which can be decomposed into two identical triangles with height 7 cm and base 14 cm.

$\therefore$ Area of zone $A = 2\left(\frac{1}{2}\times 14\times 7\right)\text{cm}^2$

$= 98\text{ cm}^2$

$\therefore$ $P(\text{zone } A) = \frac{98\text{ cm}^2}{1764\text{ cm}^2}$

$= \underline{\underline{\frac{1}{18}}}$

$\because$ Radius of the circle $= \frac{28}{2}\text{ cm} = 14\text{ cm}$

$\therefore$ Area of zone $B = (\pi\times 14^2 - 98)\text{ cm}^2$

$= \left(\frac{22}{7}\times 14^2 - 98\right)\text{cm}^2$

$= 518\text{ cm}^2$

$\therefore$ $P(\text{zone } B) = \frac{518\text{ cm}^2}{1764\text{ cm}^2}$

$= \underline{\underline{\frac{37}{126}}}$

Area of zone $C = (1764 - 518 - 98)\text{ cm}^2$

$= 1148\text{ cm}^2$

$\therefore$ $P(\text{zone } C) = \frac{1148\text{ cm}^2}{1764\text{ cm}^2}$

$= \underline{\underline{\frac{41}{63}}}$

(b) Expected value of the marks

$= 9\times\frac{1}{18} + 3\times\frac{37}{126} - 2\times\frac{41}{63}$

$= \underline{\underline{\frac{5}{63}}}$